15. AREA

- Area of a parallelogram :-

Area of a parallelogram = Base x hight

Practice set 15.1

1. If base of a parallelogram is 18 cm and its height is 11 cm, find its area.

→ Given :- Base of a parallelogram is 18 cm.
Height of a parallelogram is 11 cm.

∴ Area of a parallelogram = Base x height
= 18 x 11
= 198

Base = 18 cm

Height = 11 cm

∴ A = B x H
= 18 x 11
= 198 cm^2

2. If area of a parallelogram is 29.6 sq cm and its base is 8 cm, find its height.

→ Area = 29.6 cm^2, Base = 8 cm

A = B x H

29.6 = 8 x H

$H = \frac{29.6}{8}$

$\boxed{H = 3.7}$ cm

3. Area of a parallelogram is 83.2 sq cm. If its height is 6.4 cm, find the length of its base.

→ Area = 83.2 sq cm , Hight = 6.4

$A = B \times H$

$83.2 = B \times 6.4$

$B = \frac{\cancel{83.2}}{\cancel{6.4}}$ $\boxed{B = 13}$

★ Area of a Rhombus :-

Important properties of Rhombus :-

1] All sides of Rhombus are equal.

2] The diagonals of Rhombus are perpendicular bisector of each other.

Ⓐ Area of Rhombus = 4 × Area of one Δ formed by diagonal.

Ⓑ Area of Rhombus = $\frac{1}{2}$ × product of length of diagonal.

Practice Set 15.2

1. Lengths of the ~~digonal~~ diagonals of a rhombus are 15 cm and 24 cm , find its Area.

→ $d_1 = 15$ cm , $d_2 = 24$ cm

Area of Rhombus = $\frac{1}{2}$ × product of lengths of diagonals

$= \frac{1}{2} \times d_1 \times d_2$

$= \frac{1}{2} \times 15 \times 24$

64 $= \frac{15 \times 24}{2}$

$= \cancel{\frac{360}{2}}$

$\boxed{= 180}$

2] Lengths of the diagonals of a rhombus are 16.5 cm and 14.2 cm, find its area.

→ $d_1 = 16.5$, $d_2 = 14.2$

Area of Rhombus $= \frac{1}{2} \times$ product of lengths of diagonal

$= \frac{1}{2} \times d_1 \times d_2$

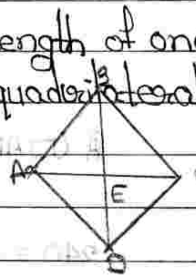
$= \frac{1}{2} \times 16.5 \times 14.2$

$= \frac{16.5 \times 14.2}{\cancel{2}}$

$= 16.5 \times 7.1$

$= 177.15 \text{ cm}^2$

3] If perimeter of a rhombus is 100 cm and length of one diagonal is 48 cm, what is the area of the quadrilateral?



→ Perimeter of Rhombus = 4 x side

$100 = 4 \times x$

$x = \frac{100}{4}$

$\boxed{x = 25}$

Diagonal of Rhombus $= BD \times \frac{1}{2} = BE = ED$

$\cancel{48} \times \frac{1}{\cancel{2}} = \boxed{24 = BE}$

Using Pythagoras Theorem

In $\triangle$ AEB

$(AB)^2 = (AE)^2 + (EB)^2$

$(25)^2 = x^2 + (24)^2$

$625 = (AE)^2 + 576$

$(AE)^2 = 625 - 576$

$(AE)^2 = 49$

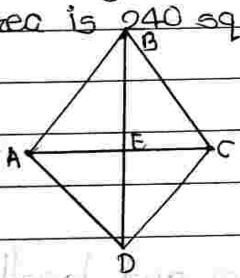
$\boxed{AE = 7cm}$ So, $\cancel{BD}$ $EC = 2 \times AE = 14$

Area of rhombus $= \frac{1}{2} \times$ product of length of diagonal

$\frac{1}{2} \times 14 \times 24$

$\boxed{= 336}$

4.* If length of a diagonal of a rhombus is 30 cm and its area is 240 sq cm, find its perimeter.



$AB = BC = CD = AD$

Let, $l(BD) = 30$ cm

$BE = DE = 15$

$A(\square ABCD) = 240\ cm^2$ (Given)

$A(\square ABCD) = \frac{1}{2} \times BD \times AC$

$240 = \frac{1}{2} \times 30 \times AC$

$240 = \frac{1}{2} \times 30 \times AC$

$480 = 30 \times AC$

$AC = \frac{480}{30}$

$\boxed{AC = 16}$

By property of triangle

$AE = CE = 8 cm$

Now, In right angle Δ AEB

$AB^2 = AE^2 + BE^2$

$= 8^2 + 15^2$

$= 64 + 225$

$AB^2 = 289$

Taking sq root

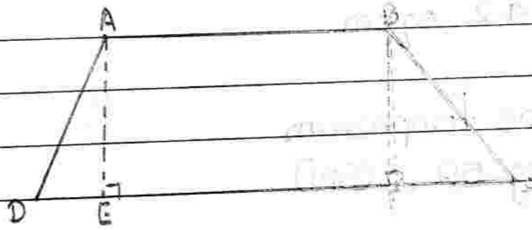
$AB = 17$

perimeter $= 4 \times AB$

$= 4 \times 17$

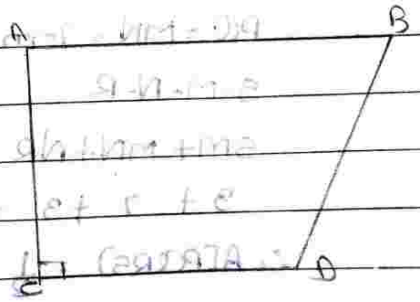
$p = 68$ cm

<u>Area of a trapezium</u> =



∴ Area of Trapezium = $\frac{1}{2} \times$ [Sum of parallel sides] $\times$ Height

Practice Set 15.3



1. In □ ABCD, l AB = 13 cm, l DC = 9 cm, l (AD) = 8 cm, find area of □ ABCD.

→ A(□ ABCD) = $\frac{1}{2} \times (AB + CD) \times AD$

$= \frac{1}{2} \times (13 + 9) \times 8$

$= \cancel{\frac{1}{2}} \times \cancel{22} \times 8$

$= 1 \times 11 \times 8$

$= 11 \times 8$

$A(\square ABCD) = 88\ cm^2$

2. Length of the two parallel sides of a trapezium are 8.5 and 11.5 cm respectively and its height is 4.2 cm, find its area.

→ Area of Trapesium $= \frac{1}{2} \times (8.5 + 11.5) \times 4.2$

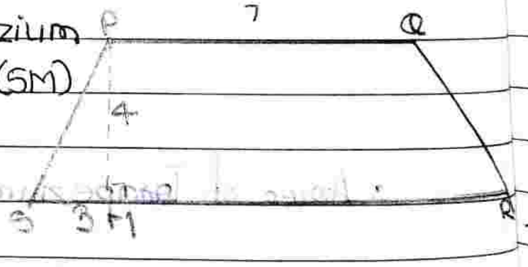
$= \frac{1}{2} \times (20.0) \times 4.2$

$= \cancel{\frac{1}{2}} \times \cancel{20} \times 4.2$

$= 1 \times 10 \times 4.2$

A of Trapezium $= 42$ sq cm

3.* $\square$ PQRS is an isosceles trapezium $l(PQ) = 7$ cm. seg PM $\perp$ seg SR, $l(SM) = 3$ cm,



Distance between two parallel sides is 4 cm, find the area of $\square$ PQRS

→ In $\square$ PQRS

PQ = MN = 7cm

S-M-N-R

SM + MN + NR = SR

3 + 7 + 3 = 13

$\therefore A[PQRS] = \frac{1}{2} \times [PQ + SR] \times PM$

$= \frac{1}{2} \times [7+13] \times 4$

$= \frac{1}{2} \times 20 \times 4$

$= \frac{1}{2} \times 80$

$= \frac{1}{\cancel{2}} \times \cancel{80}$

A[PQRS] = 40

Area of a Triangle

1] If base (b) and height (h) is given.
Then, Area of a triangle $= \frac{1}{2} \times b \times h$

2] If three sides a, b, c are given,
Then,
Area of a triangle $= \sqrt{s(s-a)(s-b)(s-c)}$
This formula is called Hero's formula.

Here s = semiperimeter
i.e. $s = \frac{a+b+c}{2}$

→ Practice Set 15.4

1] Sides of a triangle are cm 45 cm, 39 cm and 42 cm, find its area.
→ let,
a = 45 cm, b = 39 cm, c = 42 cm

$s = \frac{a+b+c}{2}$

$s = \frac{45+39+42}{2}$

$s = \frac{126}{2}$

$\boxed{s = 63 \text{ cm}}$

Now,

By Hero's formula.

$\therefore$ Area of a triangle $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{63(63-45)(63-39)(63-42)}$

$= \sqrt{63(18)(24)(21)}$

$= \sqrt{21 \times 3 \times 6 \times 3 \times 6 \times 4 \times 21}$

$= 21^2 \times 3^2 \times 6^2 \times 2^2$

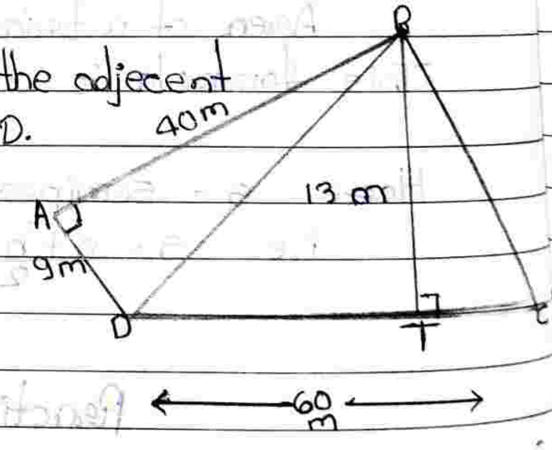
$= 21 \times 3 \times 6 \times 2$

$= 63 \times 6 \times 2$

$= 378 \times 2$

Area of a triangle $= 756 \text{ cm}^2$

3. Some measures are given in the adjecent figure, find the area of $\square$ ABCD.



$\rightarrow$ $A(\Delta BAD) = \frac{1}{2} \times AB \times AD$

$= \frac{1}{\cancel{2}} \times \cancel{40} \times 9$

$= 20 \times 9$

$A(\Delta BAD) = 180 \text{ cm}^2$ —— ①

$A(\Delta BDC) = \frac{1}{2} \times DC \times BT$

$= \frac{1}{\cancel{2}} \times \cancel{60} \times 13$

$= 30 \times 13$

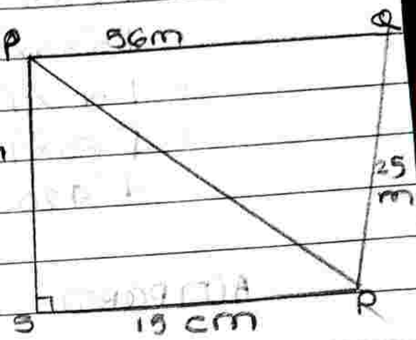
$A(\triangle BDC) = 390$ —— ②

$A(\square ABCD) = A(\triangle BAD) + A(\triangle BAC)$

$= 180 + 390$ [from ① & ②]

$A(\square ABCD) = 570$

2] Look at the measures shown in the adjacent figure and find the area of $\square$ PQRS.



→ $A(\triangle PSR) = \frac{1}{2} \times PS \times SR$

$= \frac{1}{\cancel{2}} \times \cancel{36} \times 15$

$= \frac{1}{1} \times 18 \times 15$

$A(\triangle PSR) = 270$ —— ①

By pythagoras them

$[PR]^2 = [PS]^2 + [SR]^2$

$= [36]^2 + [15]^2$

$= 1296 + 225$

$[PR]^2 = 1521$

$[PR]^2 = 39$

$A(PQR) = s = \frac{a+b+c}{2}$

$s = \frac{39+56+25}{2}$

$s = \frac{120}{6}$

$\boxed{s = 60}$

$$S = \sqrt{s(s-a)(s-b)(s-c)}$$
$$= \sqrt{60(60-39)(60-56)(60-25)}$$
$$= \sqrt{60(21)(4)(35)}$$
$$= \sqrt{10 \times 6 \times 7 \times 3 \times 2 \times 2 \times 7 \times 5}$$
$$= \sqrt{5 \times 2 \times 3 \times 2 \times 7 \times 3 \times 2 \times 2 \times 7 \times 5}$$
$$= \sqrt{5^2 \times 3^2 \times 2^2 \times 7^2 \times 2^2}$$
$$= \sqrt{5 \times 3 \times 2 \times 7 \times 2}$$
$$= \sqrt{30 \times 7 \times 2}$$
$$= \sqrt{60 \times 7}$$
$$= \sqrt{420} \quad \text{——— (2)}$$

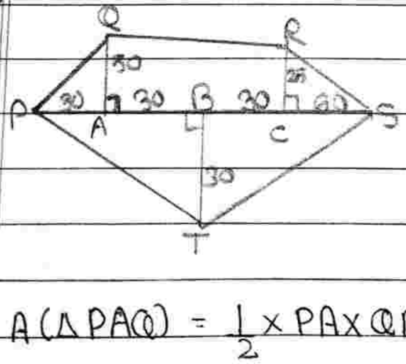
$A(\square PQRS) = A(\triangle PSR) + A(PQR)$
$= 270 + 420$
$A(\square PQRS) = 690 \text{ cm}^2$

* Area of figures having irregular shape :-

Practice Set 15.5

1. Find the areas of given plots. [All measures are in metres]

(1)



$\rightarrow A(\triangle PAQ) = \frac{1}{2} \times PA \times QA$

$= \frac{1}{2} \times 30 \times 60$

$= \frac{1}{2} \times \cancel{1500}$

$A(\triangle PAQ) = 750 \text{ m}^2$ ------- ①

$A(\square AQRC) = \frac{1}{2} \times [QA + RC] \times [AB + BC]$

$= \frac{1}{2} \times [50 + 25] \times [30 + 30]$

$= \frac{1}{2} \times [75] \times 60$

$= \frac{1}{2} \times \cancel{4500}$

$A(\square AQRS) = 2250 \text{ m}^2$ ----- ②

$A(\triangle RCS) = \frac{1}{2} \times CS \times RC$

$= \frac{1}{2} \times 60 \times 25$

$= \frac{1}{2} \times \cancel{1500}$

$A(\triangle RCS) = 750$

$A(\triangle PTS) = \frac{1}{2} \times [PA + AB + BC + CS] \times BT$

$= \frac{1}{2} \times [30 + 30 + 30 + 60] \times 30$

$= \frac{1}{2} \times 150 \times 30$

$= \frac{1}{2} \times \cancel{4500}$

$A(\triangle PTS) = 2250 \text{ m}^2$ ----- ④

$A(\square PQRST) = ① + ② + ③ + ④$

$= 750 + 2250 + 750 + 2250$

$= 150 + 4500$

$A(\square PQRST) = 6000 \, m^2$

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* Area of a circle :-

If 'r' is the radius of a circle.
Then,

① The circumference of a circle $(c) = 2\pi r$

② Area of a Circle $= \pi r^2$

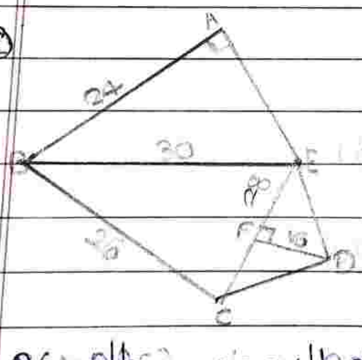
Here,

$\pi = \frac{22}{7}$ or 3.14

H.W.

Practice Set 15.2

②



→ By pythagoras thm

$(BE)^2 = (BA)^2 + (AE)^2$

$(30)^2 = (AE)^2 + 24^2$

$900 = \quad + 576$

$(AE)^2 = 900-576$

$(AE)^2 = 324$

$(AE)^2 = 18$

$A(\Delta BAE) = \frac{1}{2} \times (BA) \times (AE)$

$= \frac{1}{\cancel{2}} \times \cancel{24} \times 18$

$= 12 \times 18$

$A(\Delta BAE) = 216 m$

$A(\Delta BCE) = s = \frac{a+b+c}{2} = \frac{30+28+26}{2} = \frac{\cancel{84}}{\cancel{2}} = 42$

$s = \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{42(42-30)(42-28)(42-26)}$

$= \sqrt{42(12)(14)(16)}$

$= \sqrt{6 \times 7 \times 3 \times 4 \times 7 \times 2 \times 8 \times 2}$

$= \sqrt{3 \times 2 \times 7 \times 3 \times 2 \times 2 \times 7 \times 2 \times 2 \times 2 \times 2 \times 2}$

$= \sqrt{3^2 \times 7^2 \times 2^2 \times 2^2 \times 2^2 \times 2^2}$

$= \sqrt{3 \times 7 \times 2 \times 2 \times 2 \times 2}$

$= \sqrt{42 \times 2 \times 2 \times 2}$

$= \sqrt{84 \times 2 \times 2}$

$= \sqrt{336}$

$A(\Delta BCE) = 336 m$

$A(\Delta CDE) = \frac{1}{2} \times b \times h$

$= \frac{1}{\cancel{2}} \times \cancel{28} \times 16$

$= 14 \times 16$

$(A\Delta CDE) = 224$

$A[\square ABCDE) = A(\triangle BAE) + A(\triangle BCE) + A(\triangle CDE)$

$= 216 + 336 + 224$

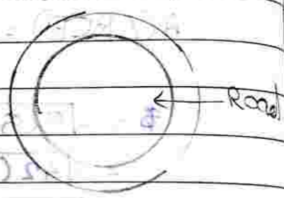
$A[\square ABCDE] = 776\ m^2$

Area of Circle

Practice Set 15.6

3. Diameter of the circular garden is 42 m. There is a 3.5 m wide road around the garden. Find the area of the road.

→ Diameter of circular garden = (d) = 42 cm



$(r) = \frac{42}{2}$

$(r) = 21$ cm

∴ Area of garden $= \pi r^2$

$= \frac{22}{\cancel{7}} \times \cancel{21} \times 21$

$= 22 \times 3 \times 21$

$= 22 \times 63$

$= 1386$ sq.m — (1)

Diameter of big circle $= 3.5 \times 42 \times 3.5$

$= 49$ m

$r = \frac{49}{2}$

Area of big circle $= \pi r^2$

$= \frac{22}{\cancel{7}} \times \frac{\cancel{49}}{2} \times \frac{49}{2}$

$= \frac{\cancel{22} \times 7 \times 49}{\cancel{2} \times 2}$

$= \frac{11\times7\times49}{2}$

$= \frac{77\times49}{2}$

$= \frac{3773}{2}$

$= 1886.5\ m^2$ — ②

Area of the road = ② - ①

$= 1886.5 - 1386$

$= 500.5$ sq.m

4. Find the area of the circle if circumference is 88 cm.

→ Circumference = $2\pi r$

$\cancel{88} = 2\times\frac{22}{7}\times r$

$616 = 2\times22\times r$

$r = \frac{616}{2\times22}$

$r = \frac{308}{22}$

$\boxed{r = 14\text{ cm}}$

Area of circle = πr^2

$= \frac{22}{7}\times14\times14$

$= 22\times2\times14$

$= 44\times14$

A of circle = $616\ cm^2$

1. Radii of the circles are given below, find their areas.

① 28 cm

→ A of circle = πr^2

② 10.5 cm

→ A of circle = πr^2

$\frac{22}{\cancel{7}} \times \cancel{28} \times 28$

$= 22 \times 4 \times 28$

$= 88 \times 28$

$\boxed{= 2464 \text{ sq cm}}$

$\frac{22}{\cancel{7}} \times \cancel{10.5} \times 10.5$

$22 \times 1.5 \times 10.5$

22×15.75

$\boxed{= 346.5 \text{ sq cm}}$

(3) 17.5 cm

→ A of circle $= \pi r^2$

$= \frac{22}{\cancel{7}} \times \cancel{17.5} \times 17.5$

$= 22 \times 2.5 \times 17.5$

$= 22 \times 43.75$

$\boxed{= 962.5 \text{ sq cm}}$

2. Areas of some circles are given below find their diameters.

(1) 176 sq cm

→ A of circle $= \pi r^2$

$176 = \frac{22}{7} \times r^2$

$r^2 = \frac{\cancel{176} \times 7}{{}_2\cancel{22}}$

$r^2 = \frac{\cancel{16} \times 7}{2}$

$r^2 = 8 \times 7$

$r^2 = \sqrt{56}$

radius × 2 = diameter

$2 \times \sqrt{56}$ = diameter